

Brand & Design System

Demo 2 — v2.0 Accessibility · WCAG 2.2 AA target Tokens · canonical in this document

!!! abstract "What this document covers" The Brand & Design System for the **Gesture-Based Drone Control System (GBDCS)**. By Demo 2 the system has progressed from wireframes to a working implementation, and the visual language has matured accordingly. This document is both a **brand guide** (palette, logo, voice) and a **design system** (tokens, components, layout, accessibility). It is served as a first-class page on the project documentation site and is the canonical source for visual decisions across the implementation.

!!! note "Demo-2 alignment" Several deliverable expectations from the Demo 2 brief land on this document. Items still in flight at the time of writing are tagged (*planned for Demo 2*). They include the standalone deployed style-guide page alongside the production app, the additional logo variants beyond the primary mark, and the self-hosting of the font payload.

1. Brand Story

GBDCS sits at the intersection of accessibility and aviation: it lets anyone fly a drone with the wave of a hand. The brand has to land in that intersection — **disciplined enough to be trusted around real flight hardware, approachable enough to feel like the first thing a new operator wants to try.**

Brand pillar	Translation into design
Accessible by default	High-contrast palette anchored on red against off-black; sans-serif typography at generous body sizes; WCAG 2.2 AA on every surface.
Real-time and alive	Motion is used to confirm system liveness — pulses on the gesture indicator, fade-in on telemetry frames — never as decoration.
Confident, not flashy	Single dominant colour (red), minimal gradients, no drop shadows on the logo, no extraneous chrome.
Operator-first	Information density tuned for a pilot who is also gesturing — large hit targets, peripheral-vision-friendly alert colours, no modals interrupting flight.

2. Colour Palette

The palette is anchored on a red core (the action colour), an off-black surface, and a small set of neutrals. Accent semantic colours are added only for **success, warning, info** and **error** states.

2.1 Core palette

Role	Token	HEX	RGB	HSL	Usage
Primary	Red	#A4161A	164, 22, 26	358°, 76%, 36%	Primary buttons, links, active states, brand surfaces.
Primary (hover)	LightRed	#BA181B	186, 24, 27	359°, 77%, 41%	Hover and pressed states on primary actions.
Primary (pressed)	DarkRed	#660708	102, 7, 8	359°, 87%, 21%	Pressed state on primary; never used as a fill colour for body areas.
Surface (dark)	OffBlack	#161A1D	22, 26, 29	206°, 14%, 10%	Light-mode page text, hero sections, dashboard chrome.
Surface (light)	OffWhite	#F5F3F4	245, 243, 244	330°, 14%, 96%	Dark-mode page text.
Muted	DarkGrey	#B1A7A6	177, 167, 166	5°, 6%, 67%	Placeholders, borders, disabled, captions.
Hairline	Grey	#D3D3D3	211, 211, 211	0°, 0%, 83%	Dividers, table rules in light mode.

How colours are referenced in code: Colours live in `tailwind.config.js` under `theme.extend.colors` and are used via the Tailwind utility classes (e.g. `bg-Red`, `text-OffWhite`). Dark mode is toggled via the `dark` class on the root element (`darkMode: 'class' in tailwind config`).

2.2 Semantic palette

Role	Token	HEX	Usage
Success	#1B7F3A	Confirmation toasts, telemetry "OK" pill, take-off complete.	
Warning	#C77700	Battery 15–25 %, link flapping, soft failsafe.	
Error	#A4161A	Critical alerts, validation errors, emergency-stop activation. (Re-uses primary red on purpose — failure is a system-level state.)	
Info	#1F6FB3	Tutorial overlays, help menu highlights.	

2.3 WCAG 2.2 contrast ratios

All foreground / background pairings used in the production UI are designed to meet WCAG 2.2 AA at a minimum, with AAA achieved for body text on both themes. The values below are the design-time ratios; an automated audit (§8.6) verifies them per build.

Foreground	Background	Ratio	Level	Use
#F5F3F4 text	#161A1D surface	15.8 : 1	AAA	Body text, dark mode.
#161A1D text	#F5F3F4 surface	15.8 : 1	AAA	Body text, light mode.
#F5F3F4 text	#A4161A primary	5.1 : 1	AA	Button labels.

Foreground	Background	Ratio	Level	Use
#F5F3F4 text	#BA181B primary-hover	4.6 : 1	AA	Button labels on hover.
#A4161A link	#F5F3F4 surface	6.6 : 1	AA	Inline links, light mode.
#BA181B link	#161A1D surface	4.7 : 1	AA	Inline links, dark mode.
#B1A7A6 muted	#161A1D surface	4.5 : 1	AA	Captions, placeholders, dark mode.
#F5F3F4 text	#1B7F3A success	4.8 : 1	AA	Success pill.
#161A1D text	#C77700 warning	6.4 : 1	AA	Warning toast (text on amber).

2.4 Usage rules

- **Red is the action colour.** Reserve it for primary buttons, links, and active states. Don't use it for decoration.
- **Off-black is the surface colour.** Pair with off-white text (never pure white — #FFFFFF on #161A1D is harsh).
- **Muted grey carries hierarchy.** Captions, placeholders, and disabled states use `DarkGrey`; avoid using it for body text.
- **Semantic colours are reserved for state.** A green pill always means "OK"; an amber pill always means "warning". Never use them for emphasis.

3. Typography

The system uses a two-family stack — a humanist sans for UI and body text, and a geometric monospace for numbers, telemetry readouts, and code blocks.

3.1 Families

Role	Family	Fallback stack	Source	Licence
UI / body	Inter	system-ui, "Segoe UI", Roboto, sans-serif	font-sans	Google Fonts (self-hosting <i>planned for Demo 2</i>) SIL Open Font Licence 1.1
Display	Geist	Inter, sans-serif	font-display	Google Fonts (self-hosting <i>planned for Demo 2</i>) SIL OFL 1.1
Monospace	System mono	ui-monospace, Consolas, monospace	-	System stack -

Self-hosting (rather than the Google Fonts CDN) is planned for the Demo 2 sprint and aligns with the SRS R8.2 posture — no runtime telemetry to external services. Until that lands, the fonts are loaded from the Google Fonts CDN at build time.

3.2 Typographic scale

A modular scale at ratio 1.2 anchored on 16 px body.

Token	Size	Line height	Weight	Letter spacing	Use
h1	32 px / 2 rem	1.2	700	-0.02em	Page titles.
h2	24 px / 1.5 rem	1.2	700	-0.01 rem	Section Headings.
h3	20 px / 1.25 rem	1.2	700	normal	Card titles.
h4	18 px / 1.125 rem	1.2	700	normal	Group labels.
h5	16 px / 1 rem	1.2	700	normal	Minor headings.
h6	14 px / 0.875 rem	1.2	700	normal	Smallest heading level.
Body (p)	-15 px / 0.95 rem	1.6	400	normal	Default paragraph text.
small	-14 px / 0.85 rem	-	400	normal	secondary/ helper paragraph text.
Code / mono	14 px / 0.875 rem	-	-	normal	Telemetry numbers, code.

All `h1–h6` elements share weight 700 and line-height 1.2 as set in `index.css`. Individual components may override weight with Tailwind utilities (e.g. `font-semibold`) where a lighter heading better suits the context.

3.3 Weights used

400 (regular), 500 (medium), 600 (semibold), 700 (bold). The team does not ship 300, 800, or 900 — they would tempt overuse of decorative weights and increase the font payload.

4. Logo & Iconography

4.1 Logo

The Codex Merchants logo currently ships in one form, with the remaining variants planned for Demo 2.

Form	File	Use	Status
Full (colour)	assets/codex_merchants_logo.png	Default. README, docs site header, landing-page hero.	Available

Form	File	Use	Status
Monogram	<code>assets/codex_merchants_mark.png</code>	Tight spaces — favicon, browser tab, app icon, ≤ 32 px usage.	Available
Monochrome	<code>assets/codex_merchants_mono.png</code>	Print, single-colour contexts.	Available
Inverse	<code>assets/codex_merchants_inverse.png</code>	On red brand surfaces only.	Available

4.1.1 Minimum size & clear space

- Full logo: minimum **120 px wide** on screen.
- Monogram (once published): minimum **24 px wide**.
- Clear space around the logo: at least the height of the monogram on all four sides. Other elements may not enter this zone.

4.1.2 Forbidden treatments

- Stretching or skewing the logo to fit a frame.
- Recolouring outside of the four documented forms.
- Applying drop shadows, glows, or strokes.
- Placing the logo on a busy photograph without a solid backing.
- Rotating the logo at any angle other than 0°.

4.2 Iconography

The system uses **Lucide** (<https://lucide.dev/>), as the single icon library (MIT-licensed).

Property	Rule
Default stroke width	1.75 px (Lucide default is 2 px; we run lighter for visual harmony with Inter).
Default size	20 × 20 px inline; 24 × 24 px standalone in buttons; 32 × 32 px for empty-state illustrations.
Colour	<code>currentColor</code> — icons inherit the surrounding text colour.
Accessibility	Standalone icon buttons must carry an <code>aria-label</code> . Decorative icons must have <code>aria-hidden="true"</code> .

Custom illustrations (e.g. the hand-gesture vocabulary card) are drawn in the same 1.75 px stroke, off-black on off-white, to sit naturally next to Lucide icons.

5. Design Tokens

Tokens are the single source of truth for visual properties. The implementation lives in `tailwind.config.js` — this document reflects those values exactly. Drift between the guide and the code is treated as a bug.

5.1 Colour tokens

See §2 for values. Colours are defined in `tailwind.config.js` under `theme.extend.colors` and consumed via Tailwind utility classes.

Token (Tailwind key)	HEX	CSS class examples
Red	#A4161A	<code>bg-Red, text-Red, border-Red</code>
LightRed	#BA181B	<code>bg-LightRed, hover:bg-LightRed</code>
DarkRed	#660708	<code>bg-DarkRed, active:bg-DarkRed</code>
OffBlack	#161A1D	<code>bg-OffBlack, text-OffBlack</code>
OffWhite	#F5F3F4	<code>bg-OffWhite, text-OffWhite</code>
DarkGrey	#B1A7A6	<code>text-DarkGrey, border-DarkGrey</code>
Grey	#D3D3D3	<code>border-Grey, divide-Grey</code>

5.2 Spacing scale

Defined in `tailwind.config.js` under `theme.extend.spacing`. Used via Tailwind's `p-, m-, gap-` utilities.

Token	Value	Use
<code>xs</code>	0.5 rem (8 px)	Tight grouping; icon to label.
<code>sm</code>	1 rem (16 px)	Form-field internal padding, list-item gap.
<code>md</code>	1.5 rem (24 px)	Card padding, section gap.
<code>lg</code>	2 rem (32 px)	Major section divider.
<code>xl</code>	3 rem (48 px)	Landing-page section padding, hero padding.

5.3 Radius

Defined in `tailwind.config.js` under `theme.extend.borderRadius`.

Token	Value	Use
<code>sm</code>	0.375 rem (6 px)	Tags, pills.
<code>md</code>	0.5 rem (8 px)	Buttons, inputs.

Token	Value	Use
lg	0.75 rem (12 px)	Cards.
x1	1 rem (16 px)	Modals, hero surfaces.
2x1	1.5 rem (24 px)	Large panels.
3x1	2 rem (32 px)	Full-bleed surfaces.

5.4 Shadow

Defined in `tailwind.config.js` under `theme.extend.boxShadow`. The glass variant requires `backdrop-filter: blur(12px)`.

Token	Value	Use
sm	0 1px 2px 0 rgb(0 0 0 / 0.05)	Cards at rest.
md	0 4px 6px -1px rgb(0 0 0 / 0.1)	Cards on hover.
lg	0 10px 15px -3px rgb(0 0 0 / 0.1)	Modals, popovers.
x1	0 20px 25px -5px rgb(0 0 0 / 0.1)	Elevated panels.
2x1	0 25px 50px -12px rgb(0 0 0 / 0.25)	Hero cards.
glass	0 8px 32px rgba(31, 38, 135, 0.37)	Glass surfaces (light).
glass-dark	0 8px 32px rgba(0, 0, 0, 0.5)	Glass surfaces over <code>OffBlack</code> .

5.5 Motion

Motion is functional, not decorative. Defined in `tailwind.config.js` under `theme.extend.animation` and implemented via Tailwind's built-in `transition-*` and `animate-*` utilities.

Utility	Duration	Use
<code>animate-spin</code>	1 s linear	Loading spinner.
<code>animate-ping</code>	1 s	Live-status pulse.
<code>animate-pulse</code>	2 s	Gesture indicator heartbeat.
<code>animate-bounce</code>	1 s	Attention-draw on empty states.
<code>transition</code> (default)	150 ms	Hover state changes (Tailwind default).

All motion respects `prefers-reduced-motion: reduce` — the global rule in `index.css` collapses all animation and transition durations to 0.01ms when the user has requested reduced motion.

All motion respects `prefers-reduced-motion: reduce` — animations collapse to a 0 ms duration when the user has requested reduced motion. The gesture pulse on the dashboard fades to a static badge in this mode.

```
@media (prefers-reduced-motion: reduce) {
  *, *::before, *::after {
    animation-duration: 0.01ms !important;
    animation-iteration-count: 1 !important;
    transition-duration: 0.01ms !important;
  }
}
```

0.01ms rather than 0ms is intentional — it avoids a flash-of-unstyled-content on some browsers while still being imperceptibly fast.

5.6 Breakpoints

Tailwind defaults are in use (no custom breakpoints defined in `tailwind.config.js`).

Token	Value	Target
sm:	640 px	Mobile landscape, small tablets.
md:	768 px	Tablets.
lg:	1024 px	Laptops; the dashboard's primary target.
x1:	1280 px	Desktop.
2x1:	1536 px	Operator workstation, large monitors.

5.7 Backdrop blur

Defined in `tailwind.config.js` under `theme.extend.backdropBlur`.

Token	Value	Use
xs	2 px	Subtle frosting.

Token Value	Use
sm	4 px Light glass panels.
md	12 px Modal backdrops.
lg	16 px Dashboard overlays.
xl	24 px Full-screen glass.

6. Component Library

Visual specifications for every component used in the production system.

6.1 Button

Variant	Background	Text	Use
Primary	bg-Red → hover:bg-LightRed	text-OffWhite	The main action on a screen. Exactly one per view.
Secondary	bg-transperant border border-Red	text-Red	Auxiliary action next to a primary.
Ghost	bg-transperant no border	text-OffWhite	Tertiary action; toolbars.
Danger	bg-Red(same as primary)	text-OffWhite	Destructive action (emergency stop, delete session).

Sizes. sm (32 px height), md (40 px, default), lg (48 px, landing-page CTAs).

States.

- Default — flat fill at the listed colours.
- Hover — fill steps to LightRed; shadow-md.
- Focus — ring-2 ring-Red ring-opacity-40 (3 px equivalent).
- Active — fill steps to DarkRed.
- Disabled — opacity-50; cursor-not-allowed; no hover effect.
- Loading — replace label with a 16 px spinner; button width preserved; click events suppressed.

6.2 Input (text, email, password)

| Property | Value | | --- | | Height | 40 px | | Padding | px-3 (0.75 rem) horizontal | | Border | border border-DarkGrey | | Background | bg-OffWhite (light) / bg-white / 4 (dark) | | Focus ring | focus:ring-2 focus:ring-Red/40 (achieved via Tailwind, outline: none' is set globally in index.css) | | Error border | border-Red; helper text below in text-Red' |

6.3 Select / Dropdown

Same shape as the Input. Caret is a Lucide chevron-down at 16 px. Open state shows a panel with shadow-lg and rounded-md.

6.4 Modal

Property	Value
Width	up to 560 px (max-w-lg)
Backdrop	bg-OffBlack/60 backdrop-blur-md
Surface	bg-OffWhite (light) / bg-OffBlack (dark)
Radius	rounded-xl
Shadow	shadow-xl
Open duration	animate utilities / transition (respects prefers-reduced-motion)
Escape key	Closes the modal; focus returns to the trigger.

6.5 Toast

Variant	Surface	Border
Success	bg-green-600/12	border border-green-600
Warning	bg-yellow-600/12	border border-yellow-600
Error	bg-Red/12	border border-Red
Info	bg-blue-600/12	border border-blue-600

Toasts auto-dismiss after 5 s (warning) or 8 s (error). Critical alerts (link loss, battery < 15 %) **do not auto-dismiss** — they remain until acknowledged per R1.2.2.

6.6 Card

rounded-lg shadow-sm at rest, shadow-md on hover, padded with p-sm (1 rem). The telemetry panel, gesture-indicator panel, and session-list rows on the replay view are all Cards.

6.7 Gesture Indicator (domain-specific)

A 96 × 96 px circular badge in the top-right of the dashboard, filled with `bg-Red` when a gesture is locked. Pulses at `animate-pulse` when the gesture changes; static in `prefers-reduced-motion`. The current gesture name and its mapped command sit immediately below.

6.8 Telemetry Pill

A pill (`rounded-full`) carrying the link status, flight mode, or battery percentage. Background is the matching semantic colour at 12 % opacity; foreground is the matching semantic colour at full opacity.

7. Layout & Spacing

7.1 Grid

Tailwind's built-in 12-column grid (`grid grid-cols-12`) with a `'gap-md (1.5 rem / 24 px)'` gutter on screens `>=md:` and `'gap-sm' (1rem / 16 px)` below.

7.2 Responsive behaviour

Surface	< m: (768 px)	≥ md:	≥ g: (1024 px)
Landing page	single column, hero stacks vertically	two-column feature grid	three-column feature grid
Dashboard	video feed stacks above telemetry; gesture indicator floats top-right	side-by-side video and telemetry	full layout: feed + overlay + telemetry + replay timeline
Replay view	session list collapses behind a drawer	session list as left sidebar	session list + waveform + telemetry chart

7.3 Spacing rules

- Inside a card: padding is `p-sm (1 rem)`; gap between elements is `gap-xs(1.5 rem)`.
- Between cards in a grid: gap is `gap-md (1.5 rem)`.
- Page-level horizontal padding: `px-sm` on mobile, `px-lg` on tablet, `px-xl` on desktop.

8. Accessibility

GBDCS targets **WCAG 2.2 AA** as the minimum on every surface. AAA is achieved for body-text contrast on both themes.

8.1 Keyboard navigation

- Every interactive element is reachable by `Tab` in document order.
- Modals trap focus; `Esc` closes them and returns focus to the trigger.
- The emergency-stop button has a global keybinding (`Ctrl+.`) so it never depends on focus state.

8.2 Focus indicator

A 3 px `ring-2 ring-Red/40` ring on every focusable element. The ring is **never** suppressed by `outline: none` without an equivalent visual replacement.

8.3 Screen readers

- Every `aria-label` is human-readable, not a token name.
- The live gesture indicator carries `aria-live="polite"` so a screen reader announces gesture changes without interrupting.
- The critical-alert region carries `aria-live="assertive"`.
- All form fields have associated `<label>` elements; placeholders are *not* used as labels.

8.4 Motion

The system honours `prefers-reduced-motion: reduce`. All animations defined in §5.5 collapse to 0 ms; the gesture pulse becomes a static badge.

8.5 Colour

No information is conveyed by colour alone. Alerts also carry an icon and a text label; telemetry pills carry a text value alongside the colour.

8.6 Audit targets

Surface	Tool	Target	Status
Landing page	Lighthouse	Accessibility ≥ 90	<i>Audit pending</i>
Dashboard	axe DevTools	0 violations	<i>Audit pending</i>
Help menu	WAVE	0 errors, 0 contrast errors	<i>Audit pending</i>

9. Voice & Tone

GBDCS speaks to pilots. The voice is **calm, direct, and specific**.

Surface	Tone	Example
Button labels	Imperative verb, ≤ 3 words.	"Take off", "End session", "Stop now".
Empty states	Friendly, action-oriented.	"No sessions yet — start a flight to see your history here."
Success messages	Quiet confirmation.	"Drone connected." (Avoid: "☑ Successfully connected to drone!" — too loud for an operational surface.)
Errors	Cause first, then suggestion.	"Camera not detected. Connect a webcam or check your browser permissions." (cause + suggestion, per R11.3.)
Critical alerts	Telegraphic, urgent, never alarmist.	"Link lost. Hovering until link returns."

Tone is **never jokey** in the operational paths. The Help menu and landing page may be warmer, but the dashboard itself stays professional — an operator in flight does not want a system that's trying to be funny.

10. Changelog from Demo 1

Area	Demo 1	Demo 2	Rationale
Format	Brand section inside the (now-retired) <code>DESIGN.md</code>	Standalone document on the docs site; canonical reference for visual decisions	The retired <code>DESIGN.md</code> mixed brand and software-design content; splitting it gives each its own deliverable. A separately deployed live style-guide page alongside the app is <i>planned for Demo 2</i> .
Palette	7 colours (3 reds, off-black, 3 neutrals)	Same 7 + 4 semantic colours (success, warning, error, info) with full WCAG audit	The early UI lacked formal state colours; adding them gave the dashboard a vocabulary for telemetry and alerts.
Typography	"Use a sans-serif"	Inter + Inter Display + JetBrains Mono with a 1.2 modular scale and named tokens	Implementation needed concrete sizes; the modular scale eliminated ad-hoc font sizes in code.
Tokens	Implicit in code	Documented in §5 with the canonical names used by the CSS layer	Drift between docs and code was happening; making the document the canonical source makes drift visible.
Components	Wireframes only	Full component spec (button, input, select, modal, toast, card, gesture indicator, telemetry pill) with all states	The system now exists, so the components have variants and states worth documenting.
Accessibility	Not formally addressed	WCAG 2.2 AA conformance target with audit targets and per-PR automation planned	Mentor feedback after Demo 1 flagged a11y as a gap; targets and audit cadence are now explicit.
Voice & tone	Absent	Added §9 with examples per surface	Several Demo 1 error messages were unclear; making the tone rules explicit fixed the inconsistency.
Iconography	"Lucide" mentioned in passing	Stroke width, sizing rules, ARIA rules, custom-illustration alignment	The Demo 1 iconography varied by view; the rules in §4.2 enforce a single visual language.
Motion	Not addressed	Three named durations + <code>prefers-reduced-motion</code> rule	The dashboard's gesture pulse made some Demo 1 reviewers motion-sick; the reduced-motion rule resolves that.

11. Wireframes

The two operator-facing surfaces. Both realise the requirements cited in their captions.

=== "Login Page"

```
![Login Page Light and dark] (assets/WF-Login.png)
```

*Figure 11.1 - Operator Authentication (UC-4, R16.1.3). Left panel with brand imagery; right form with email, password, remember me t

=== "Signup Page"

```
![Signup Page Light and Dark] (assets/WF-Signup.png)
```

*Figure 11.2 - New user registration (R16.1.1, R16.1.2). Mirrors login layout with additional fields: first name, last name , date of

=== "Gesture Dashboard"

![Gesture Dashboard Light and Dark](assets/WF-GestureDashboard.png)

*Figure 11.3 - Live gesture control and telemetry (UC-1,R1.1.1-R1.1.3).Left sidebar with menu ;centre live video feed with hand-land

=== "Analytics Page"

![Analytics Light and Dark](assets/WF-Analytics.png)

*Figure 11.4 - Telemetry history and flight metrics (UC-2, R1.1.3). Key stats (flight time , avg speed, max altitude) at top; flight

=== "Tutorial Page"

![Tutorial Light and Dark](assets/WF-Tutorial.png)

*Figure 11.5 - User Tutorial (UC-7).central to left shows the live camera feed where users can try out the gestures that are shown in